

Final Project Report

Exploring the Latino Labor Migration in Massachusetts



Generated by PlaygroundAI

Abstract

As of 2023, 13.1% of Massachusetts' total state population (~910,000 residents) is Hispanic. This growth has been substantially influenced by labor migration, notably since the 1960s, as many Latinos sought and recognized economic opportunities within the state. This data visualization contains graphs and other data visualization tools that display the population growth over time and the social and economic impact that Latinos have made, tracing their contributions and achievements from the 20th century to the present day.

Updated May 17, 2024

Description

Introduction

Massachusetts has witnessed significant population growth, with Hispanic/Latino individuals comprising 13.1% of the total state population as of 2023. The increased Latino labor migration has caused this growth in the last couple of decades. In the 1960s, waves of Latino migrants recognized and pursued economic opportunities within Massachusetts, contributing to the state's social and economic landscape. This data visualization aims to delve deeply into the story of Latino labor migration and its impact on Massachusetts, tracing the migration trajectory from the 20th century to today. With compelling data-driven insights and visualizations, we look forward to learning about the role Latinos have played in shaping Massachusetts' diverse identity.

Motivation

Coming from Venezuela of Chinese descent, I wanted to make a data visualization representing the Latino culture. While considering whether to focus on the impact of Asians or Latinos in Massachusetts, my background and experiences as a Venezuelan led me to choose to represent the contributions of Latinos, especially in the labor aspect in the United States. Growing up in Venezuela, I have a deep connection to the Latino culture, which is why I wanted to showcase the significant impact that Latinos have made in Massachusetts, as I live in Boston.

Description

This project highlights the Hispanic/Latino impact on the state of Massachusetts. The visualization graphs and charts use data from the years 1990 to 2022. The project shows a linear graph showing the significant population growth of Latinos, an interactive map showing the Latino population in each county of MA, pie charts describing the percent increase of the immigrant population in MA and the Latino workforce, and bar graphs showing that the majority of immigrants in MA come from Latin American countries, comparing and contrasting with the other regions. With all these visualizations, I intend to show the Latino influence in MA and how it looks promising to the next generation of Latinos and immigrants, whether that be in education, work, research, or other sectors.

Datasets

1. Immigrants in Massachusetts:

<https://map.americanimmigrationcouncil.org/locations/massachusetts/>

This website shows data regarding immigrants in Massachusetts. This does not entail only Latinos, but other immigrants. I would like to show general details about the population, such as demographics between foreign-born and U.S.-born populations, data regarding the number of immigrant workers in the labor force, top occupations with the highest share of immigrant workers, the share of STEM workers who are immigrants, and many other in-depth details.

2. Hispanics and Latinos in Massachusetts:

https://en.wikipedia.org/wiki/Hispanics_and_Latinos_in_Massachusetts

Wikipedia gives us multiple sources regarding the percentage of Latino individuals in Massachusetts, by country of origin. From their table, I plan to create a bar graph to show the total population in Massachusetts per country of origin.

3. Hispanic Population in Massachusetts, by Country of Origin (2010-2022):

[https://data.census.gov/table/ACSDT1Y2022.B03001?q=b03001&q=040XX00US25,25\\$16000000](https://data.census.gov/table/ACSDT1Y2022.B03001?q=b03001&q=040XX00US25,25$16000000)

This dataset shows the Hispanic population in Massachusetts, ordered by the country of origin. I got a collection of .csv files (one per year), and I plan to create bar/line graphs to show the population growth in Massachusetts and Boston throughout the years. Also, I intend to use this data to make an interactive map to show the population growth in Massachusetts throughout the years.

4. State Immigration Data Profiles (1990-2022):

<https://www.migrationpolicy.org/data/state-profiles/state/demographics/MA>

This dataset shows demographics regarding foreign-born and U.S.-born, from 1990 to 2022. From this, I plan to show the percent change of foreign-born and U.S.-born individuals. More specifically, I'd like to emphasize the percent change of Latino immigrants.

5. The Importance of Immigrants to Boston's Continued Prosperity:

<https://www.tbf.org/-/media/tbforg/images/video-stills/forum-files/alvaro-lima-sldie-on-importance-of-immigrants.pdf?la=en&hash=69D51D806AF0DED740ECB88565B0D4C307DDF5EC>

These slides show relevant data and graphics regarding the importance of immigrants in Boston. The data goes all the way back to the late 1800s. Also, it mainly shows Boston's total population, divided into regions of origin (Asia, Africa, Europe, and the Americas), and many more graphs. The story told here is that there was an economic, social, and educational advancement from immigration, and I'd like to follow a similar approach in terms of both storytelling and graph choices.

6. Boston's Immigrant Labor Force:

<http://www.bostonplans.org/getattachment/73239d8c-07de-48ac-9f9c-a150282b8a4b>

This report compares labor force integration and standard living indicators of immigrant workers in Boston. This report will be used for more data and further comprehension if needed.

7. The State of Latino Education (2010-2020):

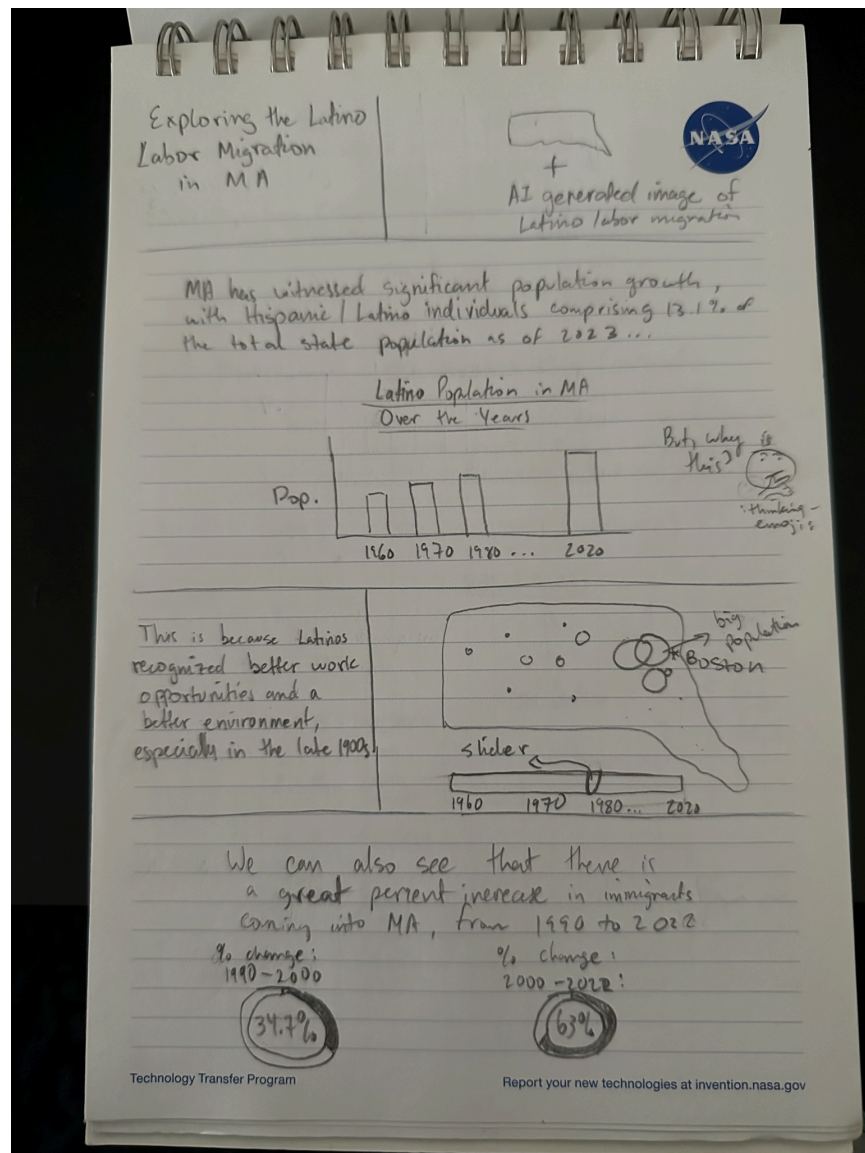
https://scholarworks.umb.edu/cgi/viewcontent.cgi?article=1293&context=gaston_pubs

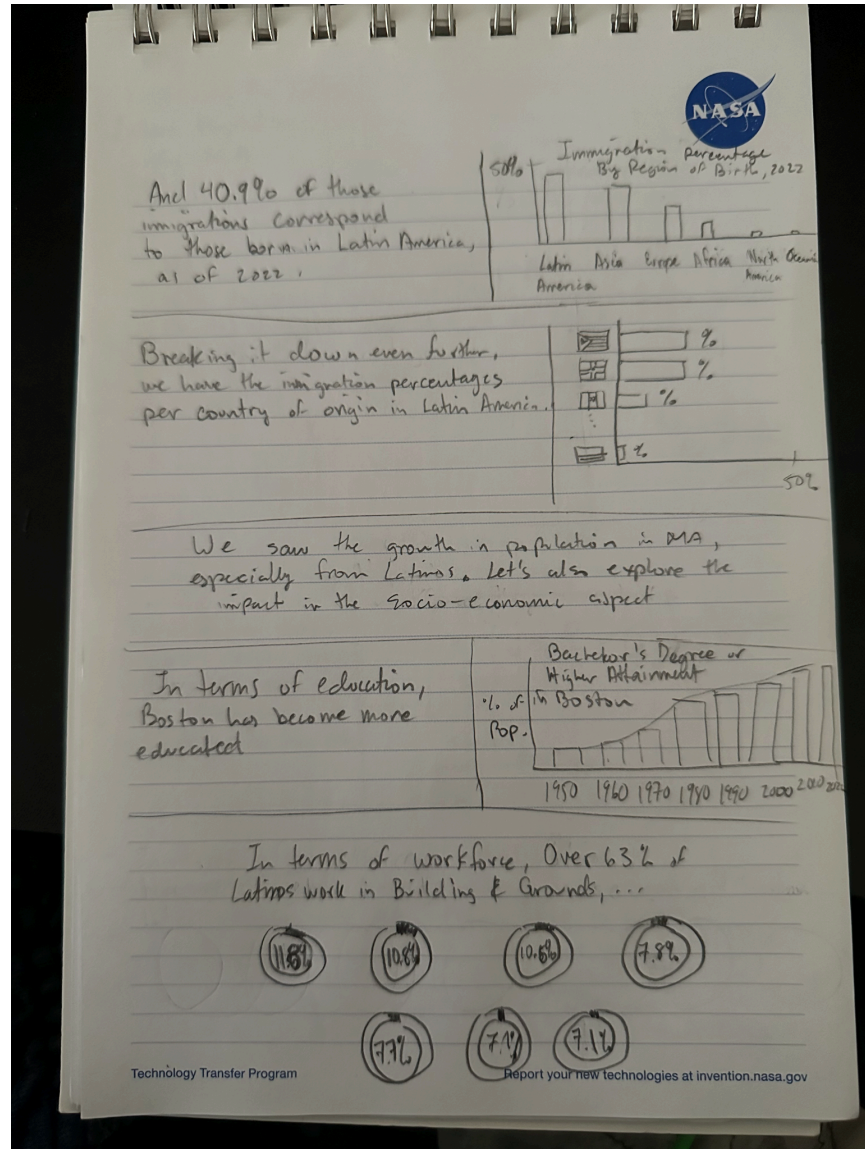
This report shows us the recent education state in Massachusetts, especially the Latino impact on education. This report will be used to explore the educational influence that Latinos have had in recent years in Massachusetts.

All the data preprocessing for the final product was done in Google Colab:

<https://colab.research.google.com/drive/1YmraXhpVxp6sx9pIZKUpjsiL3Fg18A-d?usp=sharing>

Paper Sketches







Conclusion

It's important to recognize the influence that Latinos have had in MA. There are more foreign-born and US-born Latinos in recent years, and the population is expected to grow, as well as educational attainment, success in STEM-related fields, healthcare, and many more. It's important to recognize that MA is advancing rapidly, and Boston is a living testimony of that with how culturally diverse it has become.

Sources

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- ...

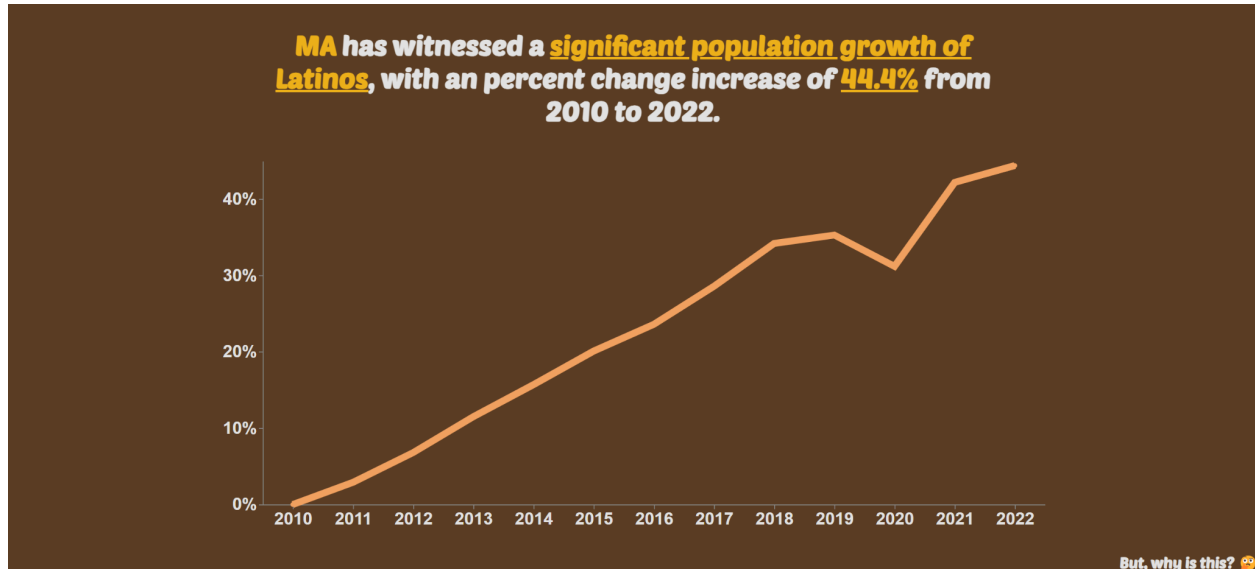
Technologies, Development, and Deployment

1. UI/UX:
 - a. Canva and Colorbrewer
2. Data Preprocessing:
 - a. Google Colab and Python
3. Data Visualization:
 - a. Structure: HTML, CSS, and JavaScript
 - b. Graph Interactivity: Plotly.js, Vega-Lite, and SVGs
 - c. Animations: GSAP
4. Website Host: GitHub Pages

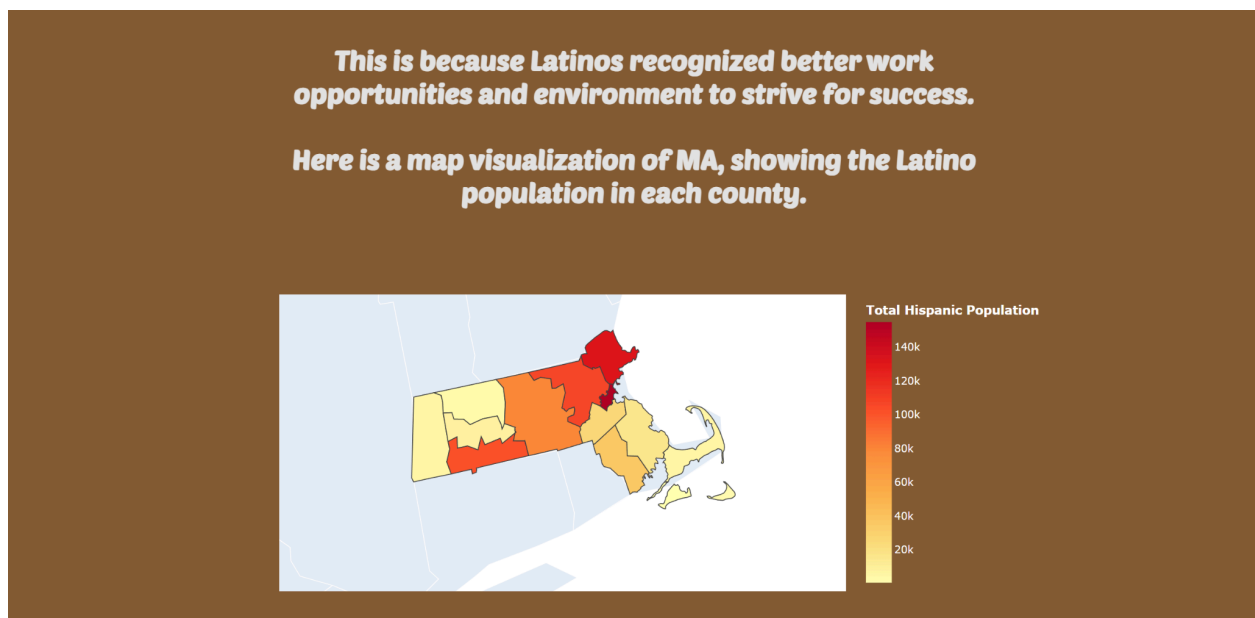
Final Product



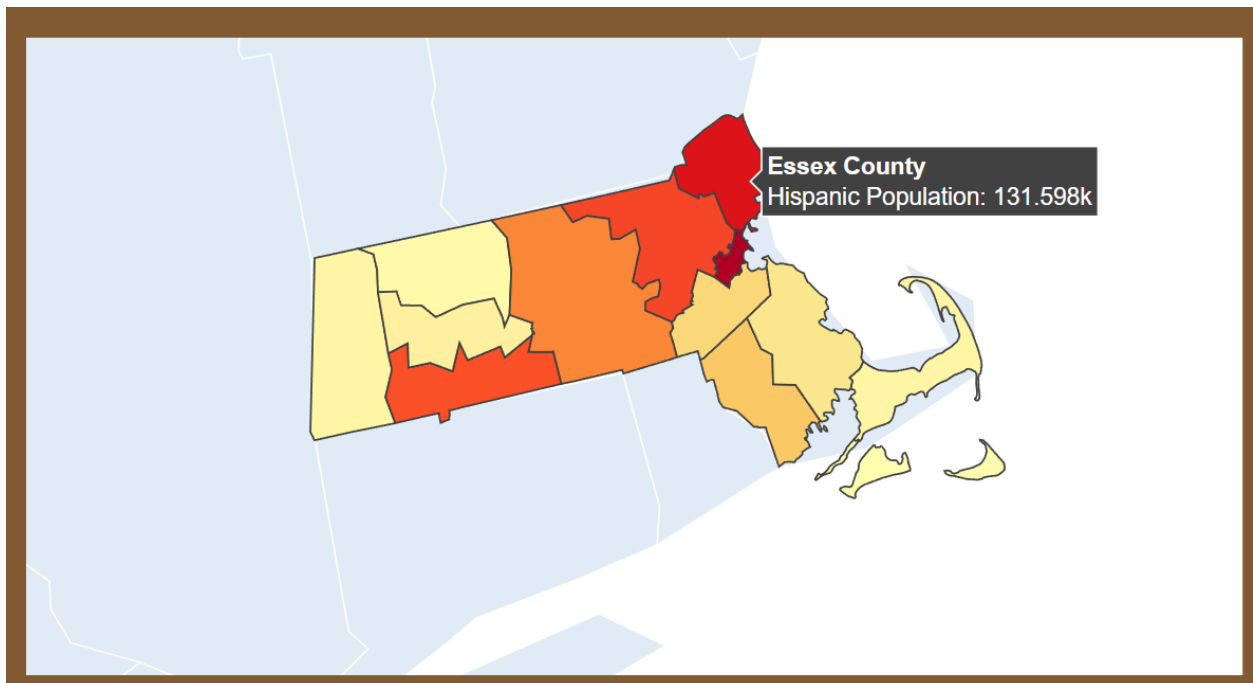
This is what the user should see once they access the website. As we can see, we have the title explaining the purpose of the website, with some images surrounding it to provide more visuals for the user to get the idea (Latin American flags + workforce + MA state).



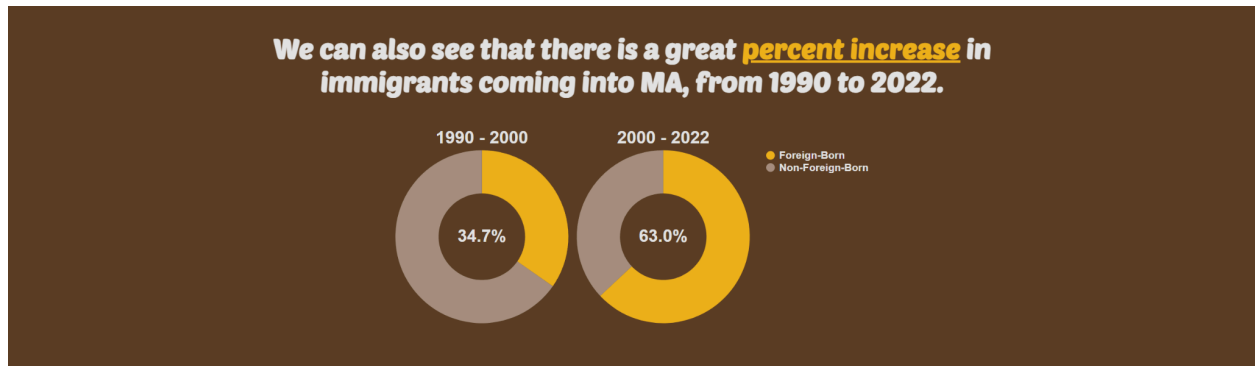
This will be the first graph visualization that the website contains, where it is shown that the Latino population is exponentially increasing as time goes on. More specifically, I showed the percent change in the Latino population from 2010 to 2022. I highlighted relevant details that I wanted the audience to get right away, especially the 44.4% increase in population within a decade, which is a huge jump. At the bottom of the graph, I asked the audience why this jump happened in MA, to which I responded in the next visualization. I used Vega-Lite to make this line plot.



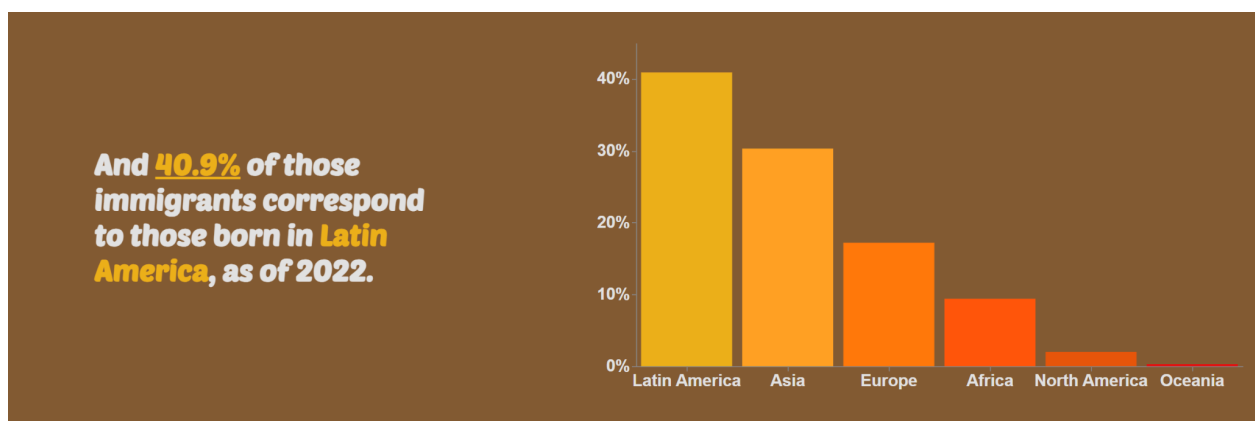
Here, I utilized Plotly.js along with layout.geo, which helps us create a map visualization with the proper API calls, given from its documentation and examples. From this visualization, I wanted to show the distribution of the Latino population in each of the MA counties. This visualization adds a different flavor to this website, and I wanted the users to interact with the map, as they can hover over each of the counties and see the total Hispanic population. The colors came from ColorBrewer, where the most intense red represents that there is a huge population and the brightest yellow represents that there is a small population.



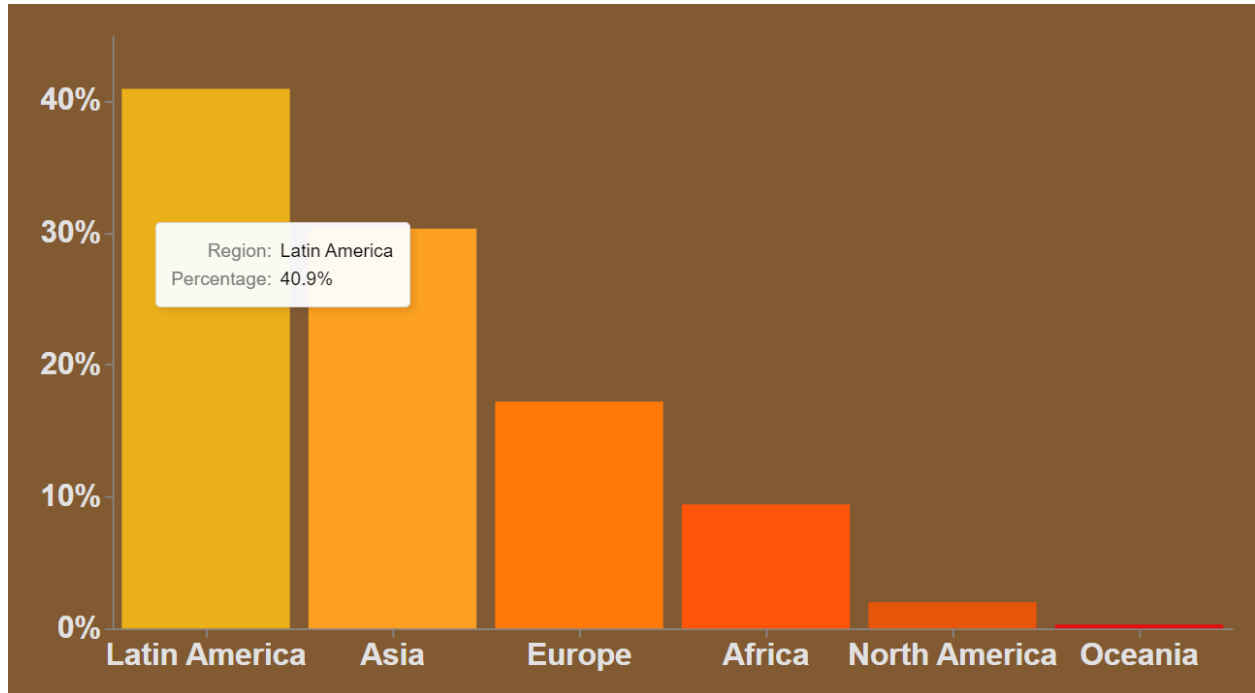
Also, when hovering over the counties, the user will be able to see the Hispanic population and the county.



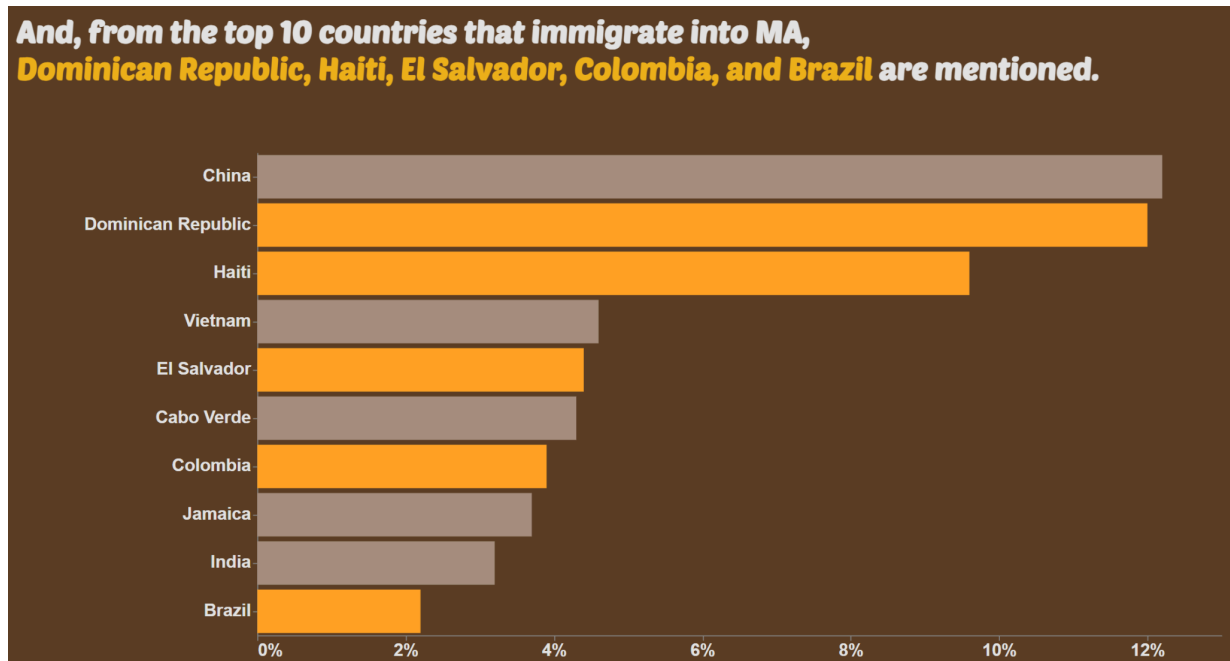
After that, I decided to show the total number of immigrants coming into MA, from 1990 to 2022. I decided to do this despite showing the first visualization because I wanted to, first, represent all immigrants, and second, to help me show the next visualizations, as they are related. I decided to show donut charts rather than pie charts because they're easier to interpret than pie charts, and I added the percent increase from the different time ranges to make it easier for the audience to get the numbers. Here, I showed the percentage between foreign-born and non-foreign-born.



Once we know the percentage of immigrants coming into MA, out of the previous percentage, I was able to show that the majority of them were Latinos. Here, I showed the other immigrants by their respective regions of origin. Here, I sorted the values from highest to lowest, moving from left to right, as I believe it's much more readable for the audience. Also, I chose a lighter color to start the trend, starting with Latin America and ending in Oceania. I had the darker color in the beginning, but it did not feel right, so I switched the order and I believe it looks better.

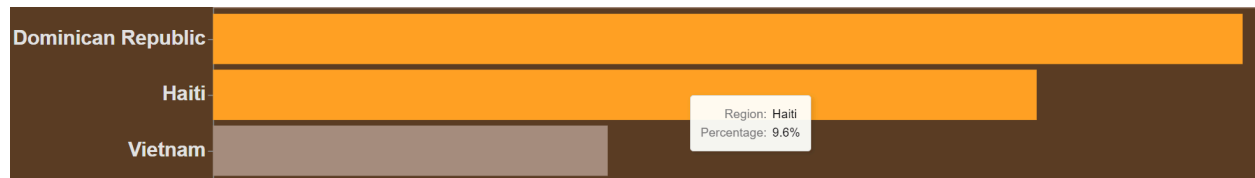


Also, when the user hovers over the bars, they are able to see the region and the percentage value. This was added for user interactivity.

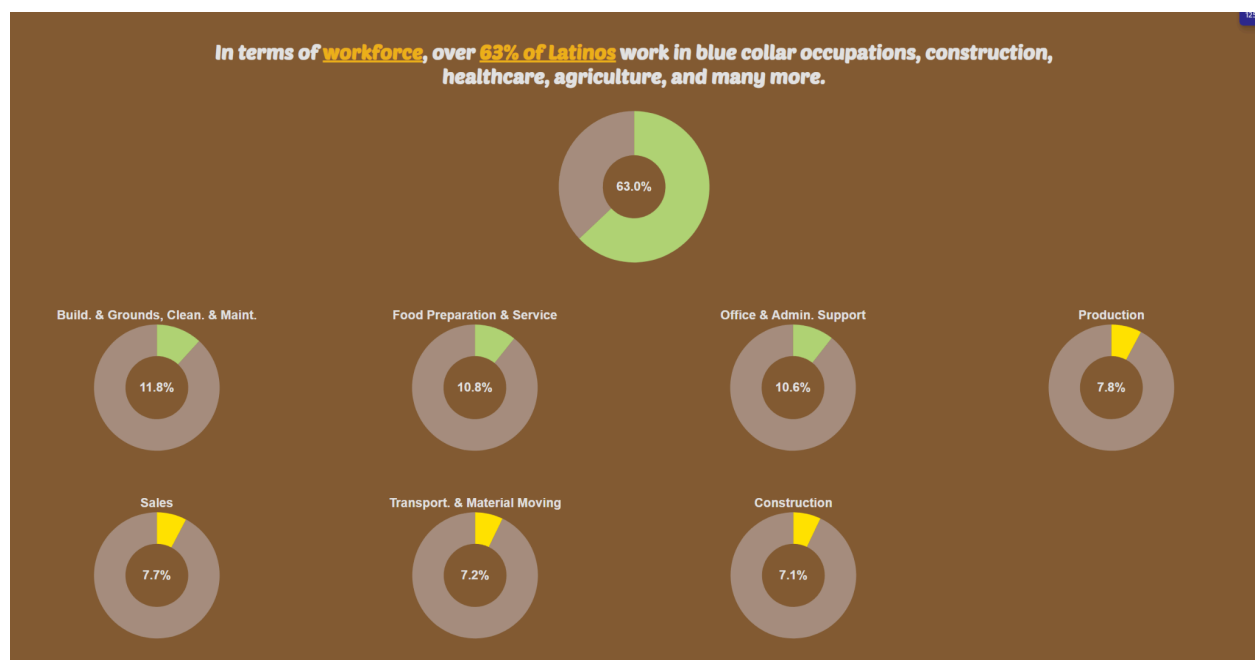


I broke it down even further by showing that, we have 5 Latin American countries out of the top 10 countries that immigrated into MA recently. Puerto Rico is also a great contributor to the Latino population, but it is a U.S. unincorporated territory. Here, I highlighted the Latin American

countries with orange, while the other countries are of a grayish color. I believe this is to get the audience's attention immediately.



Similar to the previous visualization, I added a tooltip so they can see the data when hovering over the bars.



Here, I decided to show the split of the workforce that Latinos took part in. I highlighted the fact that over 64% of Latinos work in these main 7 categories. I took inspiration from a website that did something similar for all immigrants, so I wanted to show it using data about the Latino workforce. Even though the percentages look smaller and may show that non-Latinos taking over these main categories, that is not the fact. The majority of workers in these areas are Latinos, and their percentages are much higher compared to non-Latino workers. However, the percentages needed to add up to 64% so I left it like that, avoiding complicated calculations (haha).

It's important to recognize the influence of Latinos in the state of MA. There are more foreign-born and U.S.-born Latinos in recent years, and the population is expected to continue growing, as well as educational attainment, success in STEM-related fields, healthcare, and many more important areas. It's important to recognize that MA is advancing rapidly, and Boston is a living testimony of that growth, as it has become more culturally diverse.



Sources:

- PlaygroundAI: AI-generated images
- American Immigration Council
- Census Gov.
- Migration Policy Institute
- The Importance of Immigrants to Boston's Continued Prosperity, by Alvaro Lima

Finally, I wrote a short conclusion describing the importance of the impact that Latinos have had in Massachusetts. We continue to see growth in the Latino population due to a better lifestyle and opportunities compared to their country of origin. Not only that but I explained also that the next generation of Latinos will have much more chance of striving for success in MA compared to either their or their parents' country of origin. MA has one of the best education in the U.S., and there is a great future ahead for the next generation.

Project Location

The final product can be found here:

<https://wilhenalbertohm.github.io/Labor-Migration-Vis/view/>

The GitHub repository can be found here:

<https://github.com/WilhenAlbertoHM/Labor-Migration-Vis>

References

PlaygroundAI: <https://playground.com/create>

Hispanics and Latinos in MA (Wikipedia):

https://en.wikipedia.org/wiki/Hispanics_and_Latinos_in_Massachusetts

MA Reports Data: <https://miracoalition.org/get-the-facts/massachusetts-reports-data/>

Immigrants in Massachusetts:

<https://map.americanimmigrationcouncil.org/locations/massachusetts/>

Hispanic Population in Massachusetts, by Country of Origin (2010-2022):

[https://data.census.gov/table/ACSDT1Y2022.B03001?q=b03001&q=040XX00US25,25\\$1600000](https://data.census.gov/table/ACSDT1Y2022.B03001?q=b03001&q=040XX00US25,25$1600000)

State Immigration Data Profiles:

<https://www.migrationpolicy.org/data/state-profiles/state/demographics/MA>

Immigrant Demographics:

<https://www.boston.gov/departments/immigrant-advancement/immigrant-demographics>

The Importance of Immigrants to Boston's Continued Prosperity:

<https://www.tbf.org/-/media/tbforg/images/video-stills/forum-files/alvaro-lima-sldie-on-importance-of-immigrants.pdf?la=en&hash=69D51D806AF0DED740ECB88565B0D4C307DDF5EC>